

Diagnosis and Treatment of Acute Myocarditis

A Review

Enrico Ammirati, MD, PhD; Javid J. Moslehi, MD

IMPORTANCE Acute myocarditis, defined as a sudden inflammatory injury to the myocardium, affects approximately 4 to 14 people per 100 000 each year globally and is associated with a mortality rate of approximately 1% to 7%.

OBSERVATIONS The most common causes of myocarditis are viruses, such as influenza and coronavirus; systemic autoimmune disorders, such as systemic lupus erythematosus; drugs, such as immune checkpoint inhibitors; and vaccines, including smallpox and mRNA COVID-19 vaccines. Approximately 82% to 95% of adult patients with acute myocarditis present with chest pain, while 19% to 49% present with dyspnea, and 5% to 7% with syncope. The diagnosis of myocarditis can be suggested by presenting symptoms, elevated biomarkers such as troponins, electrocardiographic changes of ST segments, and echocardiographic wall motion abnormalities or wall thickening. Cardiac magnetic resonance imaging or endomyocardial biopsy are required for definitive diagnosis. Treatment depends on acuity, severity, clinical presentation, and etiology. Approximately 75% of patients admitted with myocarditis have an uncomplicated course, with a mortality rate of approximately 0%. In contrast, acute myocarditis that is complicated by acute heart failure or ventricular arrhythmias is associated with a 12% rate of either in-hospital mortality or need for heart transplant. Approximately 2% to 9% of patients have hemodynamic instability, characterized by inability to maintain adequate end-organ perfusion, and require inotropic agents, or mechanical circulatory devices, such as extracorporeal life support, to facilitate functional recovery. These patients have an approximately 28% rate of mortality or heart transplant at 60 days. Immunosuppression (eg, corticosteroids) is appropriate for patients who have myocarditis characterized by eosinophilic or giant cell myocardial infiltrations or due to systemic autoimmune disorders. However, the specific immune cells that should be targeted to improve outcomes in patients with myocarditis remain unclear.

CONCLUSIONS AND RELEVANCE Acute myocarditis affects approximately 4 to 14 per 100 000 people per year. First-line therapy depends on acuity, severity, clinical presentation, and etiology and includes supportive care. While corticosteroids are often used for specific forms of myocarditis (eg, eosinophilic or giant cell infiltrations), this practice is based on anecdotal evidence, and randomized clinical trials of optimal therapeutic interventions for acute myocarditis are needed.

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 Supplemental content

Author Affiliations: De Gasperi Cardio Center, Transplant Center, Niguarda Hospital, Milano, Italy (Ammirati); Department of Health Sciences, University of Milano-Bicocca, Monza, Italy (Ammirati); Section of Cardio-oncology & Immunology, Cardiovascular Research Institute, University of California San Francisco School of Medicine, San Francisco (Moslehi).

Corresponding Author: Javid J. Moslehi, MD, University of California San Francisco, Smith Cardiovascular Research Building, 555 S Mission Bay Blvd, San Francisco, CA 94143 (Javid.moslehi@ucsf.edu).

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Myocarditis, defined as inflammatory injury of the myocardium that can involve the cardiac conduction system and pericardial layers,^{1,2} affects approximately 4 to 14 people per 100 000 per year globally.³⁻⁵ Acute myocarditis is typically characterized by acute onset of chest pain, dyspnea, palpitations, and occasionally syncope and the presence of myocardial inflammatory infiltrates in the setting of nonischemic cardiomyocyte necrosis on an endomyocardial biopsy (EMB) or autopsy.⁶ As a noninvasive alternative to identify myocardial edema, in the past decade, cardiac magnetic resonance (CMR) has been increasingly used instead of EMB to diagnose acute myocarditis (eTable in the Supplement).^{1,7-9} Causes of myocarditis include viruses, such as coronaviruses, influenza, and parvovirus B19; autoimmune disorders, such as systemic lupus erythematosus; drugs, such as

immune checkpoint inhibitors; and vaccines, such as mRNA COVID-19 vaccine or smallpox vaccine.^{1,2,10-12} Approximately 25% of patients with myocarditis have left ventricular (LV) systolic dysfunction, ventricular arrhythmias, or acute heart failure.¹³ This review summarizes current evidence regarding the diagnosis and treatment of acute myocarditis.

Methods

We searched PubMed for English-language studies on myocarditis published between January 1, 1982, and December 7, 2022, using the title keyword *myocarditis*. A total of 7283 articles were retrieved. Of 987 articles identified, 98 were included in this review,

consisting of 1 clinical trial, 12 reviews (2 systemic reviews), 2 meta-analyses, 63 observational studies (49 longitudinal studies and 14 cross-sectional studies), 1 case report, 3 autopsy studies, and 16 guidelines, scientific statements, or consensus documents.

Epidemiology

The most common cause of acute myocarditis is viral infections, and approximately 18% to 80% of patients present with a prodrome such as flu-like symptoms, respiratory tract symptoms, and gastrointestinal symptoms.¹³⁻¹⁶ In 2 registries that included 651 patients with acute myocarditis, 7% of patients had myocarditis due to an autoimmune disorder.^{13,17} Approximately 1% of patients with cancer treated with immune checkpoint inhibitors experience acute myocarditis.^{18,19}

Myocarditis and COVID-19

Of 2.5 million individuals vaccinated against COVID-19 (BNT162b2 mRNA vaccine), 54 people developed acute myocarditis (2.1 cases per 100 000 persons) based on an Israeli health care organization database.²⁰ Myocarditis after the COVID-19 BNT162b2 mRNA vaccine was more common in males aged 16 to 29 years (10.7/100 000 persons).²⁰ By comparison, after a SARS-CoV-2 diagnosis, acute myocarditis occurred in 59 to 64 per 100 000 males 12 years or older and 20 to 36 per 100 000 females 12 years or older.²¹ The risk of acute myocarditis was lower following vaccination compared with SARS-CoV-2 infection without vaccination (incidence rate ratio, 5.97 [95% CI, 4.54-7.87] in patients with SARS-CoV-2 after vaccination vs 11.14 [95% CI, 8.64-14.36] in patients without vaccination), based on an English national database that included more than 42 million people.²²

Myocarditis in Low- and Middle-Income Countries

Regional variation exists in the etiology of acute myocarditis. In parts of Africa, Central and South America, the Middle East, South Asia, Southeast Asia, and the Pacific islands, most patients with acute myocarditis are evaluated using clinical, electrocardiographic, and echocardiographic criteria, often resulting in a presumptive diagnosis of acute myocarditis. The prevalence of acute myocarditis due to specific local infectious etiology (such as acute Chagas disease and dengue) can be unclear.

Acute myocarditis is uncommon in patients with Chagas disease, which is caused by the protozoan *Trypanosoma cruzi*.²³ Among 103 patients identified in Colombia with confirmed acute symptomatic Chagas disease, common symptoms were fever, malaise, and splenomegaly and approximately 17% of patients were diagnosed with acute myocarditis.²⁴ However, in patients with chronic Chagas disease, endemic in Central and South America, chronic inflammatory cardiomyopathy can become clinically manifest several years after the initial exposure to *T. cruzi*. Progression from acute to chronic Chagas cardiomyopathy occurs at an estimated rate of 1.8% to 7% annually.²³

Dengue is a mosquito-borne disease that infects approximately 96 million people worldwide each year and is particularly common in South and Southeast Asia. Symptoms of dengue virus infection range from mild fever to shock syndrome.²⁵ Dengue-associated acute myocarditis has been reported in 4% to 7% of hospitalized patients with dengue from Pakistan and India, based on troponin increase and echocardiographic abnormalities.^{26,27}

Compared with patients with dengue infection who did not have myocarditis, myocarditis was associated with a longer hospital stay and increased mortality.^{26,27} Severe dengue is characterized by bleeding, and plasma leakage from capillaries resulting in cardiovascular shock, and manifestations that can overlap with acute myocarditis.²⁵ Postmortem examination of hearts from a series of 5 patients with fatal dengue confirmed inflammatory infiltrates consistent with myocarditis (Table 1^{1,9,11-13,17,18,20,23-38}).³⁹

Pathophysiology

Myocarditis, caused by infectious or noninfectious etiologies, begins with an immune-mediated injury to the myocardium.^{1,2,40} Subsequent immune cell activation, possibly due to autoantigen-specific mechanisms, leads to progression of inflammation contributing to pathologic remodeling of the myocardium and dysfunction.^{1,2} Pathogenic genetic variants associated with dilated or arrhythmic cardiomyopathies occur in 8% to 16% of patients with myocarditis.^{41,42} Pathogenic genes in cardiac desmosomes, eg, desmoplakin (*DSP*) gene, and in cardiac sarcomeres, eg, titin (*TTN*) gene, in acute myocarditis are associated with poorer prognosis compared with patients with acute myocarditis without these gene variants.^{38,41} Autoreactive T cells against cardiac antigens may contribute to immune checkpoint inhibitor-associated myocarditis.⁴³

Clinical Presentation

In 9 registries that included 2674 patients with myocarditis, acute myocarditis often presented at a median age between 30 and 45 years and about 60% to 80% of patients with myocarditis were men.^{1,13-16,44-49} Women with myocarditis presented at older ages (median age at onset, 42-45 years). However, women and men had a similar clinical presentation.^{50,51} Approximately 82% to 95% of patients with myocarditis reported chest pain,^{13-16,44,45} 19% to 49% reported dyspnea,^{13,45,46,49} 58% to 65% developed fever,^{13,15,44} and 5% to 7% experienced syncope.^{13,16,49} Cardiogenic shock occurred in about 3% to 9% of patients with acute myocarditis (ie, fulminant myocarditis) (Table 1^{1,4,9,12,13,17,18,20,23-38}).^{13,49}

Diagnosis of Myocarditis

Patients with acute myocarditis typically have elevated levels of troponin, and inflammatory markers such as C-reactive protein, and may have electrocardiographic changes of ST-T segments and echocardiographic wall motion abnormalities, particularly in the inferior and lateral walls. CMR with or without EMB is necessary to confirm the diagnosis (Box and Table 2¹³).^{1,52}

Factors associated with acute myocarditis include a family history of myocarditis or cardiomyopathy (eg, myocarditis associated with myopathic cardiac gene variants),³⁸ recent exposure to drugs such as clozapine or immune checkpoint inhibitors,¹⁰ vaccines (eg, mRNA COVID-19 and smallpox vaccines),^{10,22} toxic substances (eg, amphetamine, scorpion bites),⁵³ and certain infections. Myocarditis may occur in patients infected by parasites after ingesting raw meat,³⁰ in patients who traveled to areas where dengue is endemic, and in patients with Lyme disease.^{25,36}

The electrocardiogram (ECG) finding is abnormal in 62% to 96% of patients with acute myocarditis.^{13,15,41,44} ST-segment elevation similar to changes observed in acute myocardial infarction occurs in approximately 58% to 70% of patients with acute myocarditis.^{13,44}

Table 1. Characteristics of the Most Common Causes of Acute Myocarditis: Results From Cohort Studies

Type of myocarditis Classification based on clinical presentation	Median age and sex prevalence	Sample size and study design ^a	Characteristic findings	Associated conditions and risk factors	Typical clinical presentation	Outcome and frequencies of therapies based on registries
Uncomplicated acute myocarditis ¹³	33 y Male, 85%	Cohort of 325 patients Retrospective study	LVEF ≥50% on echocardiogram, no ventricular arrhythmias, hemodynamic stability	Associated autoimmune disorder such as SLE and eosinophilic granulomatosis with polyangiitis in 4.2%	Chest pain in 96.6% Dyspnea in 6.2% Prodromal symptoms in 80.1% Median LVEF of 60% on echocardiogram	No deaths at 5 y Immunosuppressive drugs used in 2.8% NSAIDs used in 67.6%
Acute myocarditis complicated by LVSD/AHF ¹³	35 y Male, 69%	Cohort of 118 patients Retrospective study	LVEF <50% on echocardiogram Signs of AHF	Associated autoimmune disorders in 15.4%	Chest pain in 59.1% Dyspnea in 55.7% Prodromal symptoms in 81.7% Median LVEF of 35% on echocardiogram	Cardiac death or heart transplant during hospitalization of 11.9% and 14.7%, respectively, at 5 y Immunosuppressive drugs used in 37.2% NSAIDs used in 44.0%
Acute myocarditis complicated by ventricular arrhythmias ²⁸	44 y Male, 77%	Cohort of 156 patients Retrospective study	Onset characterized by the presence of SVT or VF	Family history of cardiomyopathy in 8% Risk factors associated with ventricular arrhythmia recurrence: SVT at presentation (HR, 2.90); fibrosis involving ≥2 myocardial segments on CMR (HR, 4.51); and absence of edema on CMR (HR, 2.59)	SVT at presentation in 67% VF at presentation in 33% Median LVEF of 50% on echocardiogram Lymphocytic myocarditis was the most frequent histology in 89%	37.2% had a recurrence of ventricular arrhythmias after a median follow-up of 23 mo Ventricular arrhythmia recurrence occurred after a median of 8 mo
Fulminant myocarditis ²⁹	42 y Male, 51%	Cohort of 165 patients Retrospective study	Cardiogenic shock requiring inotropes or mechanical circulatory supports	Associated autoimmune disorders in 17.7% Risk factors associated with cardiac death or heart transplant: need for temporary mechanical circulatory support other than intra-aortic balloon pump (HR, 3.27) Giant cell histology (HR, 3.03) QRS interval >120 ms (HR, 1.74)	Dyspnea in 66.6% Chest pain in 37.0% Syncope in 16.6% Median LVEF of 22% on echocardiogram Lymphocytic myocarditis was the most frequent histology in 72.7%	Cardiac death or heart transplant at 60 d of 28.0% and at 7 y of 47.7% Immunosuppressive therapy used in 66.8% Use of IV corticosteroids in 20.2% and IVIG in 7.3% No association between use of immunosuppressive drugs and risk of cardiac death or heart transplant (HR, 0.78 [95% CI, 0.46-1.31])
Classification based on histology/etiology						
Lymphocytic acute myocarditis ²⁹ (presenting with LVSD/fulminant presentation)	40 y Male, 52%	Cohort of 146 patients Retrospective study	Myocardial infiltrates with T lymphocytes and macrophages Approximately 70%-75% of patients presenting with AHF/cardiogenic shock	Associated autoimmune disorders in 11% Use of immune checkpoint inhibitors in 2%	Dyspnea in 70.8% Chest pain in 38.0% Syncope in 15.5% Prodromal symptoms in 73.1% Median LVEF of 25% on echocardiogram	Transplant-free survival at 60 d of 19.5% and at 7 y of 44.5%

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Table 1. Characteristics of the Most Common Causes of Acute Myocarditis: Results From Cohort Studies (continued)

Type of myocarditis	Median age and sex prevalence	Sample size and study design ^a	Characteristic findings	Associated conditions and risk factors	Typical clinical presentation	Outcome and frequencies of therapies based on registries
Eosinophilic acute myocarditis ³⁰	41 y Male, 50%	Analysis of 179 published cases	Presence of eosinophilic cells together with lymphocytes Approximately 15%-23% of patients presenting with AHF/Cardiogenic shock	Specific drugs that can cause hypersensitivity reactions (eg, clozapine, β -lactam antibiotics, smallpox vaccine) Eosinophilic granulomatosis with polyangiitis Raw meet consumption (toxocariasis) Myeloproliferative variants of HES	Dyspnea in 59.4% Chest pain in 43.3% History of asthma in 22% Peripheral eosinophilia in 75.9%	In-hospital death in 22.3% based on histologically proven cases
Giant cell myocarditis ^{29,31-33}	43-61 y Male, 38%-50%	4 Cohorts including 126 patients 1 Prospective study (11 patients) and 3 retrospective studies	Large T-lymphocyte infiltrates, multinucleated cells (termed giant cells) and few eosinophils Approximately 3%-10% of patients presenting with AHF/Cardiogenic shock	Associated autoimmune disorders in 14%-20%	Dyspnea in 46%-78% Chest pain in 14%-21% Syncope in 22% SVT/VF in $\approx$ 14% Prodromal symptoms in $\approx$ 54% Median LVEF of 25% on echocardiogram	Transplant-free survival ranges from 11% to 73% at 1 y
Cardiac sarcoidosis ³¹	$\approx$ 51 y Female, 71%	Cohort of 351 patients Retrospective study	Presence of noncaseating granulomas and giant cells Approximately in 3% of patients presenting with AHF/Cardiogenic shock	Clinically manifest cardiac involvement occurs in about 5% of patients with pulmonary/systemic sarcoidosis The highest rates are reported in northern European and African American individuals, particularly in women ¹	HF as presenting manifestation in 15% High-degree atrioventricular block in 43% LVEF $\leq$ 50% on echocardiogram in 48% Abnormal focal cardiac uptake on FDG-PET in 86%	The 5-y estimate of survival free of aborted sudden cardiac death and heart transplant of 77% Use of corticosteroids in 99% and azathioprine in 41%
Drug-induced myocarditis						
mRNA COVID-19 vaccine-associated myocarditis ²⁰	27 y Male, 94%	Cohort of 54 patients Retrospective study	The diagnosis is increased 3-5 d after the second vaccine dose	Male sex and age between 16 and 29 y are associated with the highest incidence (10.69 cases/100 000 persons)	Chest pain in 81% Dyspnea in 6% Normal LVEF in 71% and severe LVSD in 4% on echocardiogram	Estimated mortality approximately 1% ^{20,34} Immunosuppressive drugs used in 1.9% NSAIDs used in 22.2%
Immune checkpoint inhibitor-associated acute myocarditis ^{18,19}	Median age at presentation of approximately 66 y ¹⁸ Male, 67% ¹⁸	Cohort of 122 patients Retrospective study	Generally presenting as lymphocytic myocarditis Median time from exposure to myocarditis, 30 d (IQR, 18-60) ¹⁸	The most important risk factor is combination treatment with CTLA-4 inhibitor (eg, ipilimumab) and PD-1 inhibitor (eg, nivolumab or pembrolizumab) vs monotherapy (OR, 4.3 [95% CI, 2.9-6.4]) ¹⁸	Dyspnea/fatigue in 71%-76% Chest pain in 14%-37% Myositis in 23%-30% Myasthenia-like syndrome (diplopia) in 11%-30% LVEF $\leq$ 50% on echocardiogram in 48% ¹⁹	Estimated mortality up to 50% ¹⁸
Clozapine-associated acute myocarditis ¹	38 y Male, 71%	Cohort of 503 patients with treatment-resistant schizophrenia Prospective study	Mean (SD) time from clozapine initiation to myocarditis, 15 (7) d	Incidence of 3%	Median LVEF of 64% on echocardiogram	Among patients hospitalized with myocarditis all survived Withdrawal of clozapine

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Table 1. Characteristics of the Most Common Causes of Acute Myocarditis: Results From Cohort Studies (continued)

Type of myocarditis	Median age and sex prevalence	Sample size and study design ^a	Characteristic findings	Associated conditions and risk factors	Typical clinical presentation	Outcome and frequencies of therapies based on registries
Myocarditis associated with systemic autoimmune disorders						
Acute myocarditis associated with systemic autoimmune disorders ^{13,17}	≈36 y Male, 53%	2 Cohorts of 46 patients Retrospective studies	Generally presenting lymphocytic myocardial infiltrate	≈7% of all acute myocarditis Among most frequent diagnosis: SLE, mixed connective tissue disorders, and eosinophilic granulomatosis with polyangiitis	Chest pain in 67% Dyspnea in 27% Myocarditis with pericardial effusion, lack of chest pain and high CRP levels were more often associated with systemic autoimmune disorders	Mortality rate varied based on case series and underlying systemic autoimmune disorders Use of corticosteroids varied among case series
Lupus myocarditis ³⁵	32-35 y Male, 12%	Analysis of 117 published cases	Generally presenting lymphocytic myocardial infiltrate, occasionally as giant cell myocarditis	1%-9% of patients with SLE Median SLE duration, 2.5 y at the time of myocarditis	Chest pain in 46% Dyspnea in 77% LVEF <50% on echocardiogram in 80% Increased anti-double-stranded DNA antibodies and/or low levels of complement C3/C4 in 73%	Overall long-term mortality (≈20%) Corticosteroids used in 97% Use of cyclophosphamide in 50%
Acute myocarditis induced by viruses						
COVID-19-associated acute myocarditis ¹²	38 y Male, 61%	Cohort of 54 patients Retrospective study	It can present without COVID-19 pneumonia in 57.4% Generally mild lymphocytic infiltrates	Median time between COVID-19 symptom onset and cardiac symptoms: 5 d (IQR: 2-12) Patients with COVID-19 pneumonia have higher mortality compared with those without pneumonia (15.1% vs 0%)	Chest pain in 55.5% Dyspnea in 53.7% Syncope in 5.5% Median LVEF of 40% on echocardiogram	Estimated 120-d mortality of 6.6% Use of IV corticosteroids in 46.3% and IVIG in 14.8%
Dengue-associated acute myocarditis ^{26,27}	30 y ²⁷ Male, 61%	Analysis of 1190 hospitalized patients with dengue from 2 cohorts 1 Prospective study and 1 retrospective study		Exposure to dengue viruses (DENV1-4) transmitted by <i>Aedes aegypti</i> and <i>Aedes albopictus</i> mosquitoes ²⁵ Myocarditis estimated in 4%-7% patients hospitalized with dengue fever ^{26,27}	Dyspnea in 92% Vomiting in 62% Abdominal pain in 46% Bleeding manifestations in 46%	38.5% Mortality in 13 patients with dengue complicated by acute myocarditis ²⁷
Acute myocarditis induced by other infective agents						
Acute myocarditis during acute Chagas disease ²⁴	Age ≥18 y in 66.7% of cases complicated by acute myocarditis ²⁴	Cohort of 103 patients with acute Chagas disease Retrospective cross-sectional study	When performed, most postmortem examinations of cases with acute Chagas disease showed amastigotes of <i>Trypanosoma cruzi</i> (pseudocysts) in the myocardium with severe inflammation	Endemic in South and Central America, caused by the protozoan <i>T. cruzi</i> ²³ Median time between symptoms and medical consultation: 17 d Oral transmission in 78% of cases complicated by myocarditis (consumption of contaminated food; eg, acai palm fruit and sugar cane) Vector transmission in 22% of cases complicated by myocarditis	Acute myocarditis complicated in 17% of cases of acute Chagas disease Thick smear test performed in 56% as primary diagnostic test (microscopically detectable trypomastigotes)	Mortality rate of 1%-10% during acute Chagas disease, mainly due to myocarditis ²⁴ Use of antitrypanosomal therapy in 87% (specifically, benznidazole in 73% ²⁴)

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Table 1. Characteristics of the Most Common Causes of Acute Myocarditis: Results From Cohort Studies (continued)

Type of myocarditis ^{36,37}	Median age and sex prevalence	Sample size and study design ^a	Characteristic findings	Associated conditions and risk factors	Typical clinical presentation	Outcome and frequencies of therapies based on registries
Lyme carditis	32 y ³⁷ Male, 84%	1 Review and 1 analysis of 45 published cases		Exposure to tick-borne <i>Borrelia</i> species (ie, burgdorferi) infection Carditis estimated in 0.3%–0.4% of patients with Lyme disease, appearing 1–2 mo after infection Endemic areas: Canada, some areas of the US, and Europe	In 80%–90% of patients, high-degree atrioventricular block occurs, often transient in nature In patients with Lyme carditis: Erythema migrans in 50% Syncope in 40% Fatigue in 40% Dyspnea in 33% Fever in 28% Chest pain in 20%	Presentation with fulminant myocarditis and mortality are extremely rare Temporary pacemaker needed in 72% of Lyme carditis Permanent pacemaker needed in 18% of Lyme carditis Antibiotics used in 93%
Myocarditis associated with myopathic cardiac gene variants						
Acute myocarditis associated with desmosomal gene variants ³⁸	24 y Male, 67%	Cohort of 36 patients Retrospective study	Evidence of pathogenic DSP variants in 91.7% Generally mild inflammatory infiltrate (mainly macrophages)	Family history of myocarditis in 22.2% Family history of sudden cardiac death or aborted sudden cardiac death <65 y in 16.7%	Chest pain in 91.7% Dyspnea in 8.3% Syncope in 11.1% Median LVEF of 55% on echocardiogram Septal fibrosis on initial CMR in 85.3%	Estimated 5-y risk of death, ventricular arrhythmias, or AHF of 62.5% Estimated 5-y risk of death or ventricular arrhythmias of 16.1%

^a Studies are longitudinal, unless noted otherwise in the column.

Abbreviations: AHF, acute heart failure; CMR, cardiac magnetic resonance; CRP, C-reactive protein; FDG-PET, fludeoxyglucose–positron emission tomography; HES, hypersensitivity syndrome; HR, hazard ratio; IV, intravenous; IVIG, intravenous immunoglobulin; LVEF, left ventricular ejection fraction; LVSD, left ventricular systolic dysfunction; NSAIDs, nonsteroidal anti-inflammatory drugs; OR, odds ratio; SLE, systemic lupus erythematosus; SVT, sustained ventricular tachycardia; VF, ventricular fibrillation.

Box. Commonly Asked Questions About Acute Myocarditis**How Is Acute Myocarditis Diagnosed?**

Clinical history typically consists of sudden onset of chest pain, dyspnea, fever, and occasionally syncope. Laboratory findings include elevated high-sensitivity troponin measurement, electrocardiographic changes of ST-T segments, and echocardiographic wall motion abnormalities. Cardiac magnetic resonance or endomyocardial biopsy can confirm the diagnosis of acute myocarditis.

Is an Endomyocardial Biopsy Necessary to Confirm the Diagnosis?

Patients presenting with ventricular arrhythmias, second- and third-degree atrioventricular block, acute heart failure not responding to medical therapy, or cardiogenic shock should undergo an endomyocardial biopsy to confirm the diagnosis and rule out high-risk histology, such as giant cell myocarditis. In addition, patients with immune checkpoint inhibitor-associated myocarditis may require endomyocardial biopsy to evaluate whether additional therapy with immune checkpoint inhibitor is appropriate. Most patients with myocarditis can be diagnosed with cardiac magnetic resonance.

What Are the Prevalence and Characteristics of COVID-19 and mRNA COVID-19 Vaccine-Associated Myocarditis?

The estimated incidence of acute myocarditis after mRNA COVID-19 vaccination is about 2.1 cases per 100 000 persons. Myocarditis is more common in males aged 16-29 years (10.7/100 000 persons). The risk of myocarditis is higher 3-5 days after the second vaccine dose. Most have chest pain (>80%) and mortality of 1.05%. After diagnosis of SARS-CoV-2, acute myocarditis occurred in about 59 to 64 per 100 000 males $\geq$ 12 years and 20 to 36 per 100 000 females $\geq$ 12 years.

particularly involving the inferior and lateral limb leads.⁵⁴ High-sensitivity troponin I or T levels are elevated in approximately 64% to 100% of patients.^{13,15,29} In a series of 1049 patients with acute myocarditis, C-reactive protein level was elevated in approximately 54% to 99% of patients.^{13,15,29} Serology can identify Lyme myocarditis (*Borrelia* antibodies) and myocarditis associated with HIV.¹

Echocardiogram, Magnetic Resonance Imaging, and Fludeoxyglucose-Positron Emission Tomography Scanning

On echocardiogram, changes consistent with myocarditis include increased myocardial wall thickness and abnormal myocardial echogenicity. Myocardial inflammation may contribute to wall thickness and granular echogenicity. Other observed changes are mild segmental hypokinesis typically involving the inferior and lateral walls, diastolic dysfunction, and pericardial effusion.¹ Echocardiographic global longitudinal strain was recently found to be associated with edema detected on CMR in patients with acute myocarditis.⁵⁵ Early in the course of myocarditis, LV dimensions are generally normal,^{13,44,56} and LV ejection fraction (LVEF) is preserved in approximately 75% of patients. However, cardiac function may decline rapidly during the initial days after presentation.¹³

Acute coronary syndrome is the primary alternative diagnosis that should be ruled out in patients with suspected myocarditis. Coronary angiography or coronary computed tomography (CT) is performed in about 46% to 95% of adult patients with acute myocarditis,^{13,15,44} usually in patients with cardiovascular risk factors presenting with chest pain with ST-T changes or segmental

wall motion abnormalities on echocardiogram. In younger individuals, coronary angiography/CT can also rule out coronary anatomic abnormalities or spontaneous coronary artery dissections, eg, during pregnancy or in the postpartum period.⁵⁷ While coronary CT may assist with evaluation for acute coronary syndrome, tachycardia, arrhythmias, and unstable hemodynamics can cause movement artifacts or poor-quality images. Because it is non-invasive, CMR is typically preferred over EMB. The updated 2018 Lake Louise CMR criteria for acute myocarditis are based on a combination of magnetic resonance sequences that detect extracellular edema and myocardial scar.⁸ In recent registries, EMB was performed in less than 13% of adults with possible acute myocarditis.^{13-15,44,45} Fludeoxyglucose-positron emission tomography imaging can be used to diagnose cardiac sarcoidosis with myocarditis, which is characterized by tracer uptake in the lungs, lymph nodes, and myocardium. Fludeoxyglucose-positron emission tomography can also be used to diagnose myocarditis in the setting of systemic autoimmune disease. In these patients, tracer uptake is observed in the myocardium and other affected organs, such as the aortic walls in specific vasculitides.¹

Endocardial Biopsy for Myocarditis Diagnosis

Guidelines recommend EMB as the first-line diagnostic modality (class of recommendation: I, level of evidence: B) in patients with unexplained acute cardiomyopathy complicated by hemodynamic instability, advanced degree atrioventricular block, or ventricular arrhythmias, and in patients who do not respond to medical therapy within 1 to 2 weeks (Figure 1).⁵⁸⁻⁶⁰ EMB diagnostic sensitivity is highest when the clinical suspicion of acute myocarditis is high. EMB is associated with a histologic diagnosis of myocarditis in up to 74% of patients with clinically suspected myocarditis and cardiogenic shock.⁶¹ Traditionally, EMB is performed by fluoroscopic guidance from the right ventricle in acute myocarditis.⁶² However, electroanatomic mapping-guided right- or left-sided EMB, in which EMB is guided by signals of intracardiac electrical activation in relation to anatomic location in a specific cardiac chamber, can also be performed and may be particularly helpful in patients with ventricular arrhythmias.⁶³

Histologic criteria of myocarditis, termed *Dallas criteria*, consist of the triad of myocardial immune infiltration, nonischemic myocyte death, and immune cells adjacent to dead myocytes.⁶⁴ The European Society of Cardiology criteria for myocarditis are based on the presence of at least 14/mm² CD45⁺ inflammatory cells, including at least 7/mm² CD3⁺T cells, on EMB.^{52,65} The immune cells typically consist of lymphocytes and macrophages. Histology helps categorize myocarditis into lymphocytic myocarditis, eosinophilic, giant cell myocarditis, or cardiac sarcoidosis depending on the dominant and specific cells present on biopsy.^{1,2,52,62} Lymphocytic myocarditis is the most common histologic type of myocarditis. Approximately 70% to 75% of patients with LV systolic dysfunction have myocardial infiltrates with T lymphocytes and macrophages.^{29,66} Eosinophilic myocarditis is characterized by the presence of eosinophilic cells together with lymphocytes. Giant cell myocarditis is characterized by large T-lymphocyte infiltrates, multinucleated cells (ie, giant cells), and few eosinophils. Approximately 3% of patients with acute myocarditis have a cardiac sarcoidosis, which is characterized by noncaseating granulomas and giant cells.²⁹

Table 2. Diagnostic Assessment in Adult Patients With Suspected Acute Myocarditis

Diagnostic assessment	Description
Vital signs and associated clinical signs	Assess hemodynamic status: blood pressure, heart rate, oxygen saturation, temperature, and other findings that can suggest etiology (eg, cutaneous rash, diplopia ^a).
Medical history	Check for recent respiratory infections, systemic inflammatory conditions, previous myocarditis, allergies, asthma, travels, illicit drugs, drugs including clozapine and anticancer therapies, vaccine, food (eg, raw meat consumption ^b), and family history of myocarditis, cardiomyopathy, and sudden cardiac death.
Electrocardiogram (ECG)	Approximately 62%-96% of people have an abnormal ECG. ST-segment elevation mimicking acute myocardial infarction occurs in approximately 58% of people with myocarditis. ¹³ QRS width >120 ms, second- or third-degree atrioventricular block, and ventricular arrhythmias are associated with high-risk.
Laboratory tests	Increase in troponin, creatine kinase (CK), CK-MB, C-reactive protein, differential white blood count (eg, eosinophils), hepatic and kidney function tests, arterial blood gas analysis (in patients with acute heart failure), autoimmunity screening if an autoimmune cause is suspected.
Serology	In patients with possible infectious cause, consider antibodies for HIV and <i>Borrelia</i> species.
Nasopharyngeal swab	Polymerase chain reaction to rule out SARS-CoV-2 or other respiratory viruses, such as H1N1 influenza, in patients with recent or ongoing respiratory symptoms.
Echocardiogram	Left ventricular ejection fraction is normal in about 75% of patients. Typical findings that can suggest acute myocarditis: (1) nondilated or mildly dilated left ventricle; (2) presence of increased wall thickness; (3) abnormal echogenicity; (4) generally mild segmental hypokinesia (especially in the inferior/inferolateral walls), but severe forms can present severe diffuse hypokinesia; (5) diastolic dysfunction; (6) pericardial effusion; and (7) impaired global longitudinal strain.
Coronary angiography/computed tomography angiography	May be used if it is necessary to exclude coronary abnormalities or spontaneous coronary artery dissections or coronary artery disease in patients with cardiovascular risk factors or elderly patients.
Cardiac magnetic resonance	This test may confirm the diagnosis of acute myocarditis by identifying and quantifying edema and fibrosis by specific sequences.
Endomyocardial biopsy	Indicated in patients with acute heart failure or cardiogenic shock (fulminant myocarditis), ventricular arrhythmias, and second- or third-degree atrioventricular block (Figure 1). It is the reference standard for diagnosis and allows for histologic characterization (eg, lymphocytic, eosinophilic myocarditis, giant cell myocarditis, or cardiac sarcoidosis).
Fludeoxyglucose-positron emission tomography	May evaluate for sarcoidosis or myocarditis in patients who may have systemic autoimmune disorders (ie, vasculitides). Preparation with specific high-fat, no-carbohydrate diet for at least 2 meals with a fast of 12 h prior to scan is required to suppress normal myocardial glucose utilization, thus avoiding physiological fludeoxyglucose uptake.

^a Rash can be associated with Lyme carditis or allergic eosinophilic myocarditis, while diplopia can be associated with myasthenia-like syndrome reported in immune checkpoint inhibitor-associated myocarditis.

^b The consumption of raw meat can be associated with *Toxocara canis*-related eosinophilic myocarditis.

Complications of EMB typically consist of supraventricular arrhythmias, pericardial effusion or tamponade due to perforation, and transient heart block.^{67,68} Medical centers that perform at least 35 to 40 EMBs per year report an overall complication rate of 1% or less.^{67,69}

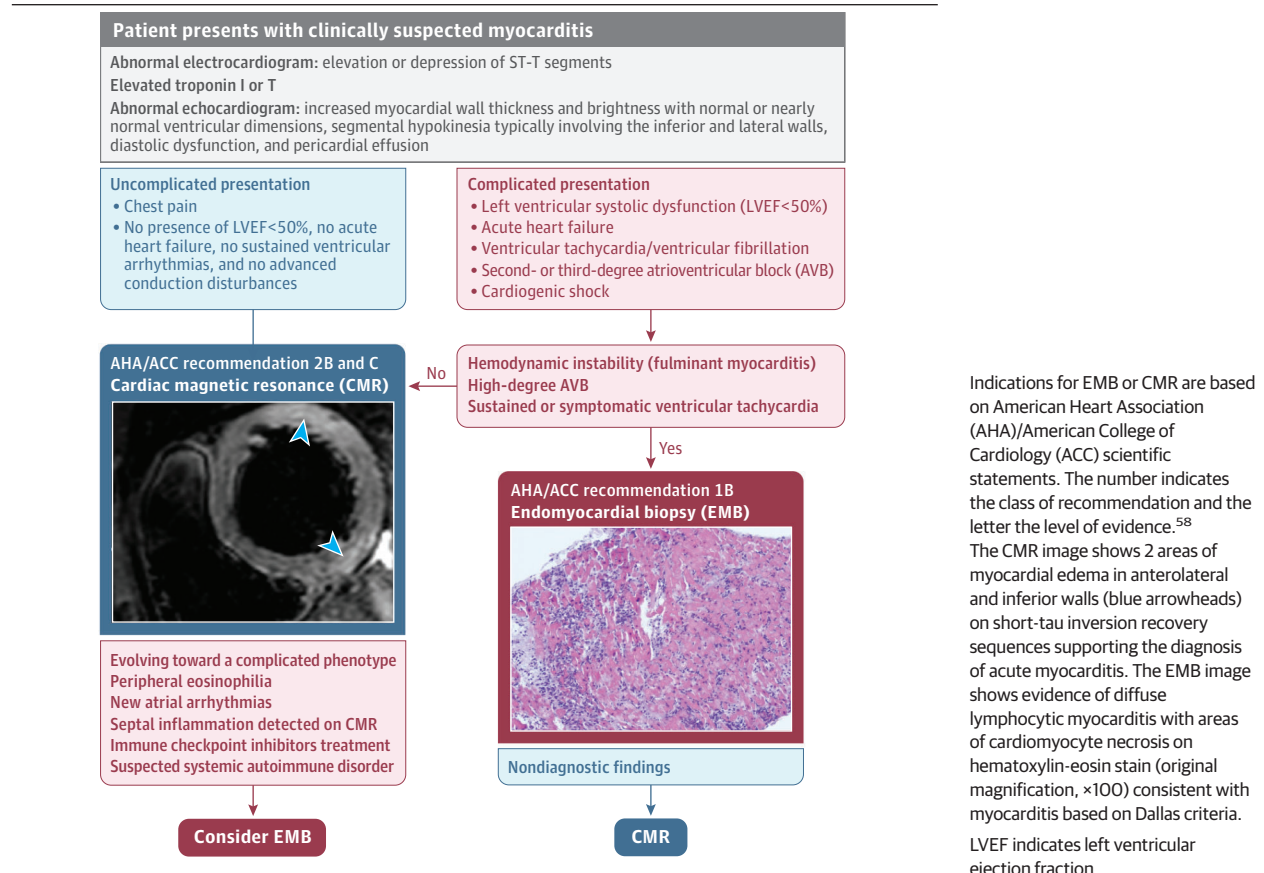
Treatment

Acute myocarditis treatment is based on acuity, severity, clinical presentation, and etiology. Patients with acute myocarditis should be categorized into complicated or uncomplicated forms of myocarditis. Patients with complicated myocarditis have LV systolic dysfunction, acute heart failure, ventricular arrhythmias, advanced atrioventricular conduction disturbance, or cardiogenic shock (Figure 2). Patients with uncomplicated myocarditis typically present with chest pain and can be treated with nonsteroidal anti-inflammatory drugs (NSAIDs), including aspirin, to relieve chest pain.^{13,44,45} In a case-control study of 45 patients, compared with patients not treated with NSAIDs, those treated with NSAIDs had no significant difference in reduction of inflammatory injury and LVEF.⁷⁰ In a longitudinal study of 181 patients with myocarditis, β -blockers were associated with better outcomes, defined as freedom from cardiac death or heart transplant, compared with absence of β -blocker treatment.⁷¹ However, the quality of evidence was poor in this observational study.⁷¹ Acute myocarditis that is complicated by conduction disturbances, consisting of second- and third-degree atrioventricular block, or ventricular arrhythmias, such as sustained ventricular tachycardia and ventricular fibrillation, may require a pacemaker, antiarrhythmic drugs (ie, amiodarone), or defibrillation. Myocarditis with ad-

vanced conduction abnormalities in patients with normal or nearly normal LVEF may have myocarditis due to Lyme disease, immune checkpoint inhibitors, or sarcoidosis.^{18,31,36,37} When the etiology of myocarditis is identified, the underlying cause should be treated (Table 3^{1,30,32,36,37,56,58,72-78}). Heart failure therapies with angiotensin-converting enzyme inhibitors, β -blockers, mineralocorticoid receptor antagonists, angiotensin receptor-neprilysin inhibitors, and sodium-glucose co-transporter 2 inhibitors are recommended in patients with reduced LVEF and stable hemodynamics.⁷⁹ Patients with severe LV systolic dysfunction generally require inotropic agents, such as norepinephrine, epinephrine, or milrinone, or temporary mechanical circulatory supports,⁵⁸ such as intra-aortic balloon pump, venoarterial extracorporeal membrane oxygenator, or intra-aortic axial pumps.^{1,58} Multicenter registries reported a short-term transplant-free survival of 66% to 76% in patients with fulminant myocarditis who received mechanical circulatory support temporarily.^{66,80} If there is no weaning from temporary mechanical circulatory supports after 2 to 3 weeks, a long-term LV assist device or urgent heart transplant may be considered.¹ The long-term outcome of patients receiving heart transplant after a diagnosis of myocarditis is similar to those of other etiologies.^{81,82} However, myocarditis recurrence has been described in some patients after heart transplant, eg, in 5% of patients with cardiac sarcoidosis and 8% of patients with giant cell myocarditis had recurrence.⁸³

Treatment of acute myocarditis with corticosteroids is controversial. To our knowledge, only 1 randomized clinical trial has assessed the efficacy of immunosuppression, compared with placebo, for acute myocarditis. In this clinical trial of 111 patients with

Figure 1. Indications for Endomyocardial Biopsy vs Cardiac Magnetic Resonance in Patients With Clinically Suspected Acute Myocarditis



myocarditis and LVEF less than 45%, prednisone combined with azathioprine or cyclosporine for 24 weeks ($n = 64$) did not significantly improve mortality compared with conventional heart failure therapy ($n = 47$).⁷³ However, treatment was administered between 2 weeks and 2 years after patients presented with acute myocarditis. Furthermore, this study was published in 1995 and may not be relevant to current practice.⁷³ No high-quality evidence supports treatment of acute myocarditis with corticosteroids.⁷²

The incidence of sudden cardiac death attributed to acute myocarditis in young adults is approximately 6% to 10%.^{1,84,85} An implantable cardioverter-defibrillator may be considered in patients at high risk of ventricular arrhythmias after myocarditis, such as those initially presenting with ventricular tachycardias or those with histologically proven cardiac sarcoidosis or giant cell myocarditis.^{28,31} High-intensity physical exercise may exacerbate myocarditis⁹ and guidelines recommend restricting competitive sports or intense physical activities for 3 to 6 months after diagnosis. However, this statement is based on expert opinion.^{52,86,87} At 6-month follow-up after acute myocarditis is diagnosed, clinicians may measure troponin level and order ECG ambulatory monitoring, echocardiogram, and a CMR to evaluate cardiac function, presence of residual myocardial edema, and size of myocardial scarring (Figure 2). A treadmill exercise test is often ordered before returning to activity, but no clinical trial evidence supports this recommendation.^{9,86,87}

Treatment of Specific Types of Myocarditis

Treating Myocarditis Due to Systemic Autoimmune Disorders

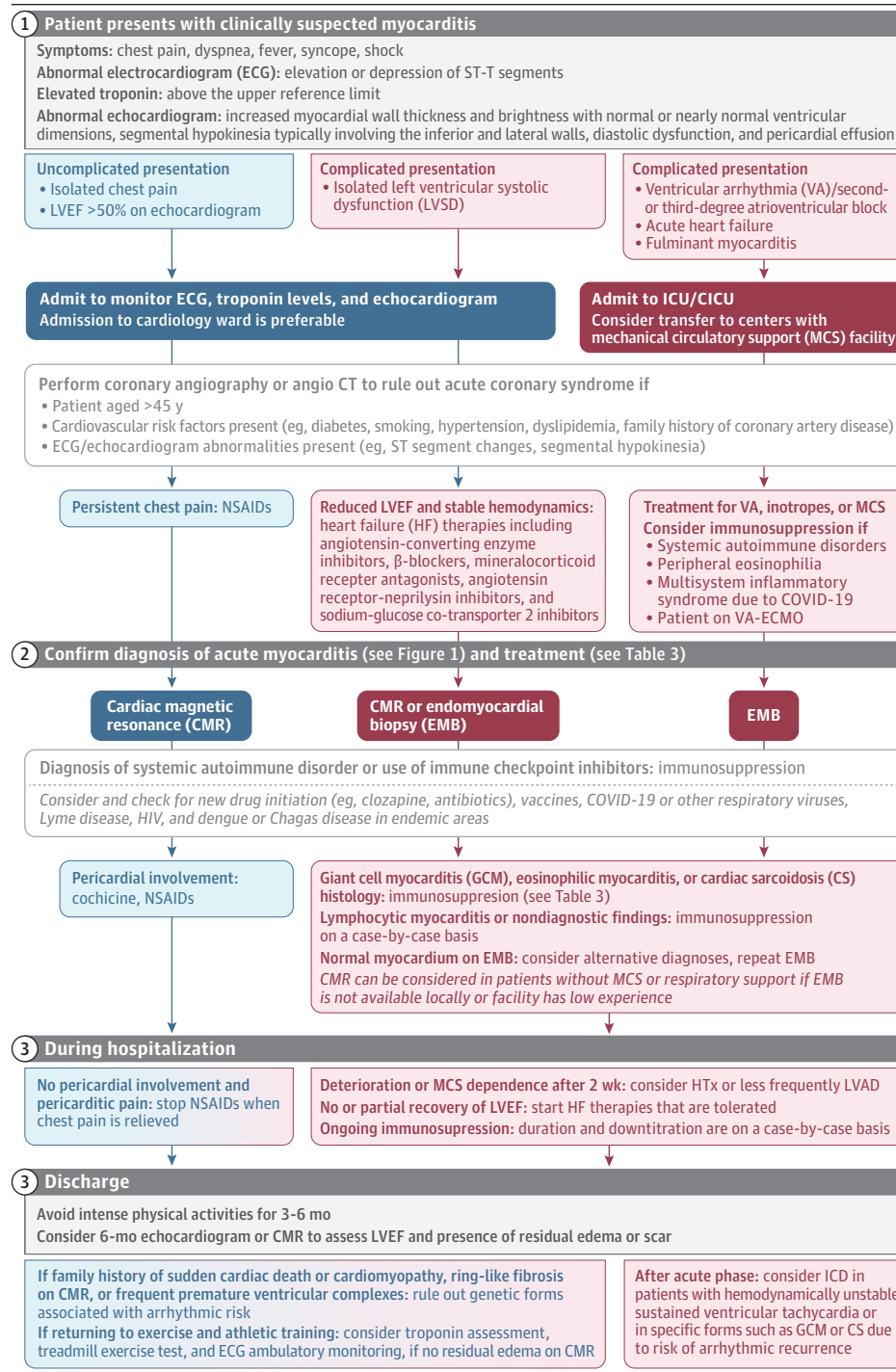
Acute myocarditis associated with systemic autoimmune disorders is generally treated with corticosteroids as first-line therapy,⁵² with higher dosages or additional immunosuppressive therapies reserved for patients with complicated presentation (Figure 2 and Table 3^{1,30,32,36,37,56,58,72-78}).^{35,74}

Treating Myocarditis Due to Immune Checkpoint Inhibitors

Immune checkpoint inhibitor-associated acute myocarditis is treated by stopping the immune checkpoint inhibitor therapy and initiating high-dose intravenous corticosteroids.⁷⁸ Additional therapies include antithymocyte globulin (anti-CD3 antibody), abatacept (a CTLA-4 agonist), and alemtuzumab (anti-CD52 antibody), but no clinical trials have demonstrated efficacy of these therapies.⁷⁸

Treating Myocarditis Due to Giant Cell Myocarditis

Giant cell myocarditis is treated with immunosuppression, including antithymocyte globulin and cyclosporine to block T cells infiltrating the myocardium combined with high-dose intravenous methylprednisolone.^{1,32,52,58} Some low-quality evidence suggests potential benefits of immunosuppressive therapy for patients with giant cell myocarditis, but no randomized clinical trials are available.³²

Figure 2. Management Algorithm of Adult Patients With Acute Myocarditis Stratified by Uncomplicated or Complicated Phenotype at Presentation

This algorithm has not been validated. CICU indicates cardiac intensive care unit; HTx, heart transplant; ICD, implantable cardioverter-defibrillator; ICU, intensive care unit; LVEF, left ventricular ejection fraction; LVAD, left ventricular assist device; NSAIDs, nonsteroidal anti-inflammatory drugs; and VA-ECMO, venoarterial extracorporeal membrane oxygenator.

Treating Myocarditis Due to Sarcoidosis

Acute myocarditis due to sarcoidosis is typically treated with corticosteroids. Methotrexate is considered second-line therapy for patients with sarcoidosis and myocarditis who do not respond to corticosteroids.¹ Other second-line or steroid-sparing therapies include azathioprine and antitumor necrosis factor inhibitors.⁷⁵

Treating Eosinophilic Myocarditis | Myocarditis with eosinophilic infiltrates is generally treated with corticosteroids based on expert opinion, but no randomized trial evidence is available. A systematic analysis of 136 published case reports of eosinophilic myocarditis not secondary to hypereosinophilic syndromes showed that corticosteroids were associated with a lower incidence of in-hospital

Table 3. Potential Therapies for Adult Patients With Acute Myocarditis

Type of myocarditis	Therapies based on cases series or recommendations based on experts' consensus ^a	Potential adverse effects of therapies (rates of adverse events)
Giant cell myocarditis ^{32,58}	Combination therapy with: IV pulses of methylprednisolone (1000 mg for 3 d) and maintenance at 1 mg/kg If hemodynamic instability: +IV ATG, 1 mg/kg, usually a single dose + oral cyclosporine twice a day (target trough levels, 150-250 ng/mL) Alternative therapy to ATG: IV alemtuzumab (anti-CD52 antibody), a single dose of 30 mg If hemodynamic stability: +oral cyclosporine twice a day (target trough levels, 150-250 ng/mL)	Methylprednisolone: hepatic function abnormalities (≥10%), malaise (≥10%), moon face (≥10%), risk of infections (≥10%), acne (≥10%), hyperglycemia (1%-10%), gastrointestinal upset (1%-10%), osteopenia (1%-10%), osteoporosis (1%-10%), insomnia (1%-10%), gastrointestinal bleeding (<1%), hypertension (<1%), febrile neutropenia (<1%), diabetes (<1%), glaucoma (<1%), and cataract (<1%)^b ATG: risk of infections (≥10%), bone marrow suppression (≥10%), febrile neutropenia (1%-10%), gastrointestinal upset (1%-10%), risk of cancer (1%-10%), rash (1%-10%), hypotension (1%-10%), serum sickness (<1%) Cyclosporine: kidney toxicity (≥10%), risk for infection,^c hypertension (≥10%), hyperlipidemia (≥10%), tremor (≥10%), hypomagnesemia (1%-10%), and hepatic function abnormalities (1%-10%) Alemtuzumab: neutropenia (≥10%), rash (≥10%), thyroid disorder ≥10%, infection (≥10%), herpes infection (≥10%), and infusion reactions (1%-10%)
Lymphocytic acute myocarditis (presenting with acute HF/fulminant presentation)	Based on case reports and case series of fulminant myocarditis^{1,56,58,72}; initial IV pulses of methylprednisolone (500-1000 mg for 3 d) and maintenance at 1 mg/kg could be considered on individual bases MTT trial showed no benefit of prednisone + azathioprine or cyclosporine in patients with lymphocytic myocarditis with LVSD⁷³ MYTHS randomized trial is assessing efficacy of IV methylprednisolone, 1000 mg, for 3 d in this setting (NCT05150704)⁷⁴	Methylprednisolone: see above
Cardiac sarcoidosis ^{1,75}	In patients presenting with recurrent VA, high-degree AVB or acute HF, IV methylprednisolone, 7-14 mg/kg, pulsed doses for 3 d, then subsequent tapering. In cases of LVSD or HFpEF, consider prednisone, 0.5 mg/kg/d Most common second-line therapy: methotrexate, 5-25 mg/wk In case of persistent/refractory cardiac inflammation consider: Oral azathioprine 1-2 mg/kg⁷⁵ IV infliximab (5 mg/kg or up to 500 mg at time 0 and after 2 and 4 wk)	Corticosteroids: see methylprednisolone adverse effects Methotrexate: hepatic function abnormalities (<1%), bone marrow suppression (<1%), gastrointestinal upset (<1%), pneumonitis (<1%), and infections^c When combined with prednisone, the patient is at risk for <i>Pneumocystis jirovecii</i> and herpes zoster Folic acid supplementation is required Azathioprine: bone marrow suppression (≥10%), gastrointestinal upset (1%-10%), hypersensitivity reaction (<1%), hepatic function abnormalities (<1%), and infections (<1%) When combined with prednisone, the patient is at risk for <i>P jirovecii</i> and herpes zoster Infliximab: viral infections (≥10%), gastrointestinal upset (≥10%), bacterial infections (1%-10%), bone marrow suppression (1%-10%), activation of tuberculosis (<1%), demyelination syndrome (<1/1000), and malignancy (<1/1000)
Eosinophilic acute myocarditis ³⁰ (hypersensitivity reaction [ie, myocarditis associated with clozapine use], eosinophilic granulomatosis with polyangiitis, raw meat consumption [toxocariasis], and myeloproliferative variant of HES)	Corticosteroids are used as first-line therapy. Type and dosage range from prednisolone, 50 mg/d, tapered over 8 wk to methylprednisolone, 1000 mg for 3 d³⁰ Withdrawal of suspected drug in case of hypersensitivity reaction ± corticosteroids (generally used in case of complicated presentation) Additional drugs are used based on the associated conditions IV cyclophosphamide 600 mg/m² (BSA) at 1, 15, and 30 d Alternatively, anti-IL-5 agents: mepolizumab, 100-300 mg SC/4 wk, or benralizumab, 30 mg SC/4-8 wk Albendazole, 600-800 mg/d, for 2-8 wk⁷⁶ Imatinib, 100-400 mg/d, for 4-28 d (up to normalization of eosinophilic count)⁷⁷	Corticosteroids: See methylprednisolone adverse effects Cyclophosphamide: bone marrow suppression (≥10%), alopecia (≥10%), urinary tract infection (≥10%), hematuria (≥10%), neutropenic fever (1%-10%), hemorrhagic cystitis (1%-10%), risk of infections (1%-10%), and infertility (1%-10%) Anti-IL-5 agents: headache (≥10%), risk of respiratory infections (1%-10%), urinary tract infections (1%-10%), hypersensitivity reactions (1%-10%), nasal congestion (1%-10%), and abdominal discomfort (1%-10%) Albendazole: hepatic function abnormalities (≥10%), gastrointestinal upset (1%-10%) headache (<1%), hypersensitivity reaction (<1%) Imatinib: bone marrow suppression (≥10%), gastrointestinal upset (≥10%), and hepatic function abnormalities (1%-10%)
Immune checkpoint inhibitor-associated acute myocarditis ^{1,78}	Withdrawal of immune checkpoint inhibitor therapy Initial IV pulses of methylprednisolone (500-1000 mg for 3 d) used in most of cases as first line Additional regimens in refractory to steroids cases based on case reports: IV abatacept (a CTLA-4 agonist), 10 mg/kg-25 mg/kg, on days 0, 5, and 12 IV ATG, 1 mg/kg, usually single dose or IV alemtuzumab (anti-CD-52 antibody), 30 mg, single dose or ruxolitinib, 10-15 mg, by mouth twice a day (usually treated for 2-4 wk)	Methylprednisolone: see above Abatacept: risk of respiratory infection (>10%), risk of urinary tract/herpetic infection (1%-10%), hypertension (1%-10%) ATG: see above Alemtuzumab: see above Ruxolitinib: risk of thrombocytopenia or anemia (>10%) and bruising (>10%)

(continued)

Table 3. Potential Therapies for Adult Patients With Acute Myocarditis (continued)

Type of myocarditis	Therapies based on cases series or recommendations based on experts' consensus ^a	Potential adverse effects of therapies (rates of adverse events)
Acute myocarditis associated with systemic autoimmune disorders ^{1,74}	Initial IV pulses of methylprednisolone (500-1000 mg for 3 d) in most of cases with presentation complicated by acute HF or ventricular arrhythmias IV cyclophosphamide, 600 mg/m ² , at days 1, 15, and 30 as most frequent additional/second-line therapy in myocarditis associated with SLE or vasculitides Initial therapy in uncomplicated presentation or initial maintenance therapy: oral prednisone, 0.5-1 mg/kg, daily Other additional/second-line therapy for maintenance: mycophenolate mofetil, 500-1500 mg, twice a day	Methylprednisolone: see above Cyclophosphamide: see above Mycophenolate mofetil: bone marrow suppression ($\geq 10\%$), gastrointestinal upset ($\geq 10\%$), hepatotoxicity (1%-10%), infections ($\geq 10\%$), and skin cancer (1%-10%) When combined with prednisone, the patient is at risk for <i>P jirovecii</i> and herpes zoster
Lyme carditis ^{36,37}	Severe cardiac presentation (ie, third-degree AVB): IV ceftriaxone, 2 g, once daily (10-14 d) Alternative regimen: oral doxycycline, 100 mg, twice a day (14-21 d) Mild cardiac presentation: oral doxycycline, 100 mg, twice a day (14-21 d) Use of temporary PM in case of third-degree AVB Permanent PM is recommended if 1:1 conduction disturbance is not restored by 14 d after admission	Ceftriaxone: bone marrow suppression (2%), eosinophilia (2%), gastrointestinal upset (2%), and rash (1%) Doxycycline: hypersensitivity reaction ($\geq 10\%$), reactivation of SLE ($\geq 10\%$), serum sickness ($\geq 10\%$), hypotension ($\geq 10\%$), angioedema, pericarditis ($\geq 10\%$), peripheral edema ($\geq 10\%$), headache ($\geq 10\%$), gastrointestinal upset ($\geq 10\%$), and photosensitivity ($\geq 10\%$)

Abbreviations: ATG, antithymocyte globulin; AVB, atrioventricular block; BSA, body surface area; CD, cluster of differentiation; CTLA-4, cytotoxic T-lymphocyte antigen-4; HES, hypereosinophilic syndrome; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; IV, intravenous; IVIG, intravenous immunoglobulin; LVSD, left ventricular systolic dysfunction; MTT, Myocarditis Treatment Trial; MYTHS, Myocarditis Therapy With Steroids; PM, pacemaker; SC, subcutaneous; and SLE, systemic lupus erythematosus; VA, ventricular arrhythmia.

^a Specific treatment efficacy based on randomized clinical trials is not available.

^b Incidence of adverse events varies significantly in acute vs chronic administration. Reported incidence of adverse events refers to chronic use.

^c Prevalence of the adverse event was unknown.

mortality compared with absence of any immunosuppressive therapy (9.9% vs 65.7%, $P < .001$).³⁰ Specific therapies other than corticosteroids are based on the associated clinical condition such as eosinophilic granulomatosis with polyangiitis,⁷⁴ helminthic infections (eg, toxocariasis),^{30,76} and myeloproliferative variants of hypereosinophilic syndromes (Tables 1 and 3).⁷⁷

Prognosis

Mortality from acute myocarditis is approximately 1% to 7%.^{3,13,88} Uncomplicated acute myocarditis is associated with in-hospital mortality rates of approximately 0%, while patients with complicated acute myocarditis experience mortality or need for heart transplant during hospitalization of approximately 12% and 15% at 5 years of follow-up.¹³ Both LV and right ventricular failure were associated with death or heart transplant among 174 patients with histologically proven acute myocarditis.⁸⁹ Fulminant myocarditis with cardiogenic shock has an estimated mortality or heart transplant at 60 days and 7 years after hospitalization of 28% and 48%, respectively, based on an international retrospective registry of 163 patients with histologically proven fulminant myocarditis.²⁹ In this registry, factors associated with increased mortality or heart transplant were giant cell myocarditis on EMB, QRS interval greater than 120 ms on initial ECG, and requirement of a temporary mechanical circulatory support other than an intra-aortic balloon pump (Table 1).²⁹ Among 156 patients with acute myocarditis presenting with life-threatening ventricular arrhythmias, characteristics associated with sudden cardiac death and ventricular arrhythmias included sustained ventricular tachycardia (78% of patients with events vs 60% of patients without events; hazard ratio, 2.90 [95% CI, 1.38-6.11]), fibrosis involving 2 or more myocardial segments (61% vs 25%; hazard ratio, 4.51 [95% CI, 2.39-8.53]), and absence of edema on initial CMR (50% vs 27%; hazard ratio, 2.59 [95% CI, 1.40-4.79]).²⁸

Myocarditis recurrence or ventricular arrhythmias occurred in approximately 3% to 9% over a follow-up period of 19 to 90 months.^{13,14,16} Among 374 patients with myocarditis, compared with those without interventricular septum fibrosis, the presence of interventricular septal fibrosis was associated with higher rates of sudden cardiac death, requirement for implantable cardioverter-defibrillator therapy, resuscitated cardiac arrest, or hospitalization for heart failure (16% vs 8%).¹⁵ In a study of 183 patients with myocarditis, myocardial fibrosis was present in 95% of patients who experienced sudden cardiac deaths vs 41% without sudden cardiac death.⁹⁰ In 97 patients with acute myocarditis, 36 patients with desmosomal gene variants (especially *DSP*) had a 62.3% risk of death, ventricular arrhythmias, and recurrent myocarditis at 5 years compared with 17.5% in those without these gene variants.³⁸ Among the 36 patients with myocarditis and desmosomal gene variants, the total number of events during follow-up was 35, including 1 death, 5 episodes of ventricular arrhythmias, and 29 episodes of recurrent myocarditis.

A study of 54 patients hospitalized with COVID-19 and definite or probable acute myocarditis had a 120-day mortality of 6.6%.¹² Mortality in patients with acute myocarditis after mRNA COVID-19 vaccine was approximately 1%.^{20,34} Mortality after immune checkpoint inhibitor-associated myocarditis was associated with a mortality rate of up to 50% in a pharmacovigilance study that included 122 cases of acute myocarditis.¹⁸

Acute Myocarditis in Children

The incidence of acute myocarditis in childhood is approximately 2 per 100 000 person-years.⁹¹ An estimated 41% to 69% of pediatric cases of acute myocarditis are preceded by a viral prodrome.⁹² Genetic variants appear to be associated with the risk of myocarditis in pediatric populations.⁹³ In 20 patients with histologically

proven myocarditis who presented dilated LV (median age, 1.4 years), 22% had pathogenic genetic variants, including *DSP* and *TTN* genes.⁹³ In 213 children in a Finnish registry who were older than 5 years, 72% to 80% of acute myocarditis was diagnosed in boys.⁹¹

The most common symptoms of acute myocarditis in children are fatigue (25%-70%), gastrointestinal symptoms (nausea/vomiting or abdominal pain; 28%-48%), and dyspnea (22%-25%). Chest pain occurs in approximately 24% to 42%.⁹² Fever precedes myocarditis in about 31% to 58% of children with myocarditis.⁹²

The diagnosis of myocarditis in children is typically made by elevated troponin levels, ECG changes, and echocardiogram results consistent with myocarditis. ECG and echocardiogram findings are similar to those observed in adults. CMR and EMB are not frequently performed.^{88,94} The differential diagnoses of acute myocarditis in children include Kawasaki disease, severe sepsis, idiopathic cardiomyopathy, primary ventricular arrhythmias, and COVID-19-related multisystem inflammatory syndrome in children. In low- and middle-income countries, the differential diagnosis includes rheumatic carditis.⁹⁵

There is no high-quality evidence to guide treatment of acute myocarditis in children. Intravenous immunoglobulins are frequently used in pediatric patients with acute myocarditis accompanied by cardiac dysfunction or ventricular arrhythmia.⁸⁸ Empirical corticosteroids are used in approximately 25% of children with myocarditis, but no clinical trial evidence supports this practice.^{88,94}

In a national registry of 898 children with myocarditis, approximately 2% to 13% died or required cardiac transplant.^{88,94} Out-

comes vary based on age at onset and the etiology of myocarditis. Mortality rates in children with myocarditis ranged from 0% among 623 adolescents and young adults in the US admitted with mRNA COVID-19-associated myocarditis (median age, 16-17 years; male prevalence, 88%-91%)^{96,97} to about 31% among 35 newborns from the US, Australia, Europe, Israel, and Hong Kong who were diagnosed with enteroviral myocarditis (median age, 7 days; male prevalence, 71%).⁹⁸

Limitations

This review has several limitations. First, the quality of included articles was not evaluated. Second, data on therapeutic interventions were primarily based on retrospective registries, case series, or case reports. Third, some relevant publications may have been missed. Fourth, myocarditis in Chagas disease, dengue-associated myocarditis, and HIV-related acute myocarditis were either not discussed or covered superficially.

Conclusions

Acute myocarditis affects approximately 4 to 14 per 100 000 people per year. First-line therapy depends on acuity, severity, clinical presentation, and etiology and includes supportive care. While corticosteroids are often used for specific forms of myocarditis (eg, eosinophilic or giant cell infiltrations), this practice is based on anecdotal evidence, and randomized clinical trials of optimal therapeutic interventions for acute myocarditis are needed.

ARTICLE INFORMATION

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Additional Information: Drs Ammirati and Moslehi contributed equally to this work.

Submissions: We encourage authors to submit papers for consideration as a Review. Please contact Mary McGrae McDermott, MD, at mdm608@northwestern.edu.

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